Two-singlet channels and the first interaction signal
A chord-space resolution of the short-time commutator

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Abstract

We resolve the previously computed singlet commutator curvature into orthogonal chord channels. Four fifths of its leading small-coupling coefficient comes from the first relative two-matter channel, whose same-species norm vanishes in the free limit; one fifth comes from a zero-matter component proportional to the Hamiltonian. The exact finite-parameter sum agrees with an independent full Wick enumeration. We also extend the positive chord generator to ordered two-matter representations and give their standard Jacobi–Laguerre limiting realization. This separates relative-channel labels from physical masses and identifies the extra operator data needed for a meson spectrum.

1 An exact decomposition of an existing observable

The state is the full annealed normalized trace τ, with $\beta = \mu_{\text{chem}} = 0$. Take complex DSSYK double scaling first, then small $h = 4\lambda$, $q = e^{-h}$, at fixed times or fixed chord degrees. Our normalized conserved-subtracted derivative singlet X has $d = 2$, $\tau X^2 = 1$, mixed crossing weight $r = q^2$, and X–X crossing weight $\chi = q^4 = r^2$. Write

$$H_s = 2\mathcal{J}_s H_{\text{src}}, \qquad \tau H_{\text{src}}^2 = h^{-1}, \qquad F = H_{\text{src}}X^2 - 2XH_{\text{src}}X + X^2 H_{\text{src}}. \tag{1}$$

Since $[\dot{X}, X] = 2i\mathcal{J}_s F$, the nonnegative commutator square is

$$\mathcal{C}(t) = \tau([X(t), X]^\dagger [X(t), X]) = 4\mathcal{J}_s^2 t^2 \|F\|^2 + O(t^4). \tag{2}$$

Our channel calculation gives the exact fixed-h decomposition

$$\boxed{\|F\|^2 = \frac{1}{h}\left[\underbrace{4(1-r)^2}_{\text{zero matter}} + \underbrace{2(1-r)^4}_{k=0 \text{ two matter}} + \underbrace{4(1-r^2)^2}_{k=1 \text{ two matter}}\right].} \tag{3}$$

Thus the respective leading coefficients in $\mathcal{C}(t)$ are $256\mathcal{J}_s^2\lambda t^2$, zero at order λ, and $1024\mathcal{J}_s^2\lambda t^2$. Their sum reproduces the earlier $1280\mathcal{J}_s^2\lambda t^2$. This resolves the existing scalar result into states; it is not a new Lyapunov exponent.

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2 Proof using the two-open-matter Gram form

Let (L, M, R) denote one Hamiltonian chord respectively before, between, or after two open X chords. The no-crossing and crossed matter pairings give

$$G = \begin{pmatrix} 1 & r & r^2 \\ r & 1 & r \\ r^2 & r & 1 \end{pmatrix}, \qquad G_\times = \chi \begin{pmatrix} 1 & r & r^2 \\ r & r^2 & r \\ r^2 & r & 1 \end{pmatrix}. \tag{4}$$

The middle crossed entry is χr^2, not χ; the Hamiltonian chord crosses both matter chords. The physical two-open-X norm is $G + G_\times$. Internal matter contractions form the separate zero-matter sector. This is the same-species Wick-sector decomposition, not the assertion that $G_\times$ is a unitary exchange matrix.

In the ordered direct form G, the normalized first primary and the symmetric external descendant are

$$u_1 = \frac{(-r, 1 + r^2, -r)^T}{\sqrt{1 - r^4}}, \qquad s_0 = (1, 0, 1)^T. \tag{5}$$

Both external annihilators kill u_1; it is the $d_{\text{eff}} = 5$ primary. Direct multiplication proves

$$\begin{gathered} u_1^T G u_1 = 1, \quad u_1^T G s_0 = u_1^T G_\times s_0 = 0, \\ u_1^T G_\times u_1 = -\chi r^2, \quad s_0^T G s_0 = 2(1 + r^2), \quad s_0^T G_\times s_0 = 2\chi(1 + r^2). \end{gathered} \tag{6}$$

The raw two-matter part of $\sqrt{h}F$ has vector $v = (1, -2, 1)^T$. Its exact projection is

$$v = c\, s_0 + a\, u_1, \qquad c = \frac{(1 - r)^2}{1 + r^2}, \qquad a = -\frac{2(1 - r^2)}{\sqrt{1 - r^4}}. \tag{7}$$

Substituting $\chi = r^2$ yields

$$\begin{aligned} \|c s_0\|^2_{G+G_\times} &= 2(1 - r)^4, \\ \|a u_1\|^2_{G+G_\times} &= a^2(1 - r^4) = 4(1 - r^2)^2. \end{aligned} \tag{8}$$

The contraction of the two X's inside $H_{\text{src}}X^2$ or $X^2 H_{\text{src}}$ leaves H_{src}; inside $X H_{\text{src}} X$ it leaves $r H_{\text{src}}$. The zero-matter part is therefore $2(1 - r)H_{\text{src}}$, with squared norm $4(1 - r)^2/h$. Orthogonality of the zero- and two-open-matter Wick sectors proves Eq. (3).

The $k = 1$ direct unit vector has physical squared norm $1 - r^4 = 1 - e^{-8h} \sim 8h$. Simultaneously $a^2 \sim 8h$. Their product divided by h gives an order-h effect in $\|F\|^2$. Dropping this channel because its unrescaled free-limit norm vanishes would discard the leading $4/5$ contribution.

3 Ordered two-matter representations and the positive generator

For dimensions $d_1, d_2 > 0$, the ordered chord states are $|l, m, r\rangle$, with $E = \bar{n} = d_1 + d_2 + l + m + r$. The coproduct gives

$$\begin{aligned} a_L|l, m, r\rangle = [l]_q|l-1, m, r\rangle + q^{d_1+l}[m]_q|l, m-1, r\rangle \\ + q^{d_1+d_2+l+m}[r]_q|l, m, r-1\rangle, \end{aligned} \tag{9}$$

and the reflected a_R action. We use the physical Gram form, whose pure-middle direct norm is $[m]_q!$. These are the existing Lin–Stanford rules [1], Sec. 3.3 and App. B.2.

Their decomposition is

$$\mathcal{R}_2^{d_1,d_2} = \bigoplus_{k=0}^{\infty} \mathcal{R}_1^{D_k}, \qquad D_k = d_1 + d_2 + k. \tag{10}$$

One may construct its normalized primary Ω_k without an all-k normalization formula. At grade k, take the orthogonal complement of the external creations from grade $k-1$. Adjointness then gives $a_L\Omega_k = a_R\Omega_k = 0$. Inductively the already constructed families fill all but one dimension, since

$$\dim \mathcal{H}_k = \binom{k+2}{2}, \qquad \sum_{j=0}^{k-1}(k-j+1) = \binom{k+2}{2} - 1. \tag{11}$$

Positivity of the chord Gram form and the one-matter irreps ensure nondegenerate descendant blocks and orthogonality of different families. This gives one new primary at each grade.

Define $K_+ = (a_L^\dagger - a_R^\dagger)/\sqrt{h}$ and $K_- = (a_L - a_R)/\sqrt{h}$ as in the preceding note, and

$$B_h = (K_+ + K_-)/2, \qquad \Pi_h = (K_- - K_+)/(2i), \qquad Q_h = E + \Pi_h. \tag{12}$$

Each summand in Eq. (10) has the preceding one-matter proof with $d = D_k$. Therefore Q_h is positive and self-adjoint on the ordered two-matter space (domain $D(E)$ at fixed h). For every fixed k, n its normalized antisymmetric descendants tend to $D^+_{D_k}$, with raising coefficient $\sqrt{(n+1)(2D_k+n)}$.

The resulting contracted sector is $\bigoplus_k D^+_{D_k}$ on finite sums of fixed-channel descendants. It is not a uniform limit of all finite-h chord directions. In particular it does not retain every singularly rescaled ordering difference automatically. The original concatenation coproduct is not a primitive sum on independent matter Hilbert spaces. Xu's factorization likewise requires a modified inner product [2].

4 The relative coordinate and why k is not a mass

The standard positive discrete-series tensor decomposition [3, 4] realizes Eq. (10) after contraction as $D^+_{d_1} \otimes D^+_{d_2}$. An explicit unitary coordinate map is

$$P = p_1 + p_2 > 0, \quad x = p_1/P \in (0,1), \qquad (\mathcal{U}f)(P,x) = \sqrt{P}\, f(Px, P(1-x)). \tag{13}$$

The measure is $dP\,dx$. Let $D = d_1 + d_2$, and define

$$\begin{aligned} \chi_k(x) &= \frac{x^{d_1-1/2}(1-x)^{d_2-1/2}}{\sqrt{N_k}} P_k^{(2d_2-1,2d_1-1)}(2x-1), \\ N_k &= \frac{\Gamma(k+2d_1)\Gamma(k+2d_2)}{(2k+2D-1)k!\Gamma(k+2D-1)}. \end{aligned} \tag{14}$$

At removable singular parameter values, take the continuous limit of the displayed normalization. The χ_k form an orthonormal basis of $L^2(0,1;dx)$. With the preceding Laguerre radial basis, the channel map is

$$e_{k,n} \longmapsto \phi_n^{D_k}(P)\chi_k(x), \qquad Q = P, \quad B = i(P\partial_P + \tfrac{1}{2}). \tag{15}$$

It intertwines E, B, Q with the sums of their one-particle realizations. In particular, if $a = d_1 - 1/2$, $b = d_2 - 1/2$, the transformed sum $S_1 + S_2$ is

$$\begin{aligned} -P\partial_P^2 - \partial_P + \frac{\mathcal{T}_x}{P}, \qquad \mathcal{T}_x &= -\partial_x x(1-x)\partial_x + \frac{a^2}{x} + \frac{b^2}{1-x} + \frac{1}{4}, \\ \mathcal{T}_x\chi_k &= (D_k - \tfrac{1}{2})^2\chi_k. \end{aligned} \tag{16}$$

This follows from the Jacobi differential equation. It verifies the tensor realization explicitly, rather than identifying a finite-h coproduct with a primitive sum.

For a free positive-energy extension with equal particle masses m and common momentum scale μ, light-front kinematics gives

$$P^+ = \mu P, \qquad P^- = \frac{m^2}{\mu P x(1-x)}, \qquad M^2 = \frac{m^2}{x(1-x)}. \tag{17}$$

At $d_1 = d_2 = 2$, direct integration in the *relative* space gives

$$\langle\chi_0|M^2/m^2|\chi_0\rangle = \frac{14}{3}, \qquad \boxed{\langle\chi_0|M^2/m^2|\chi_2\rangle = \frac{\sqrt{11}}{3} \neq 0.} \tag{18}$$

Thus even the free mass operator mixes the discrete k channels. Its spectrum is the continuum $[4m^2, \infty)$, not the list D_k. The p_i, x here parametrize the two-singlet representation; they have not been identified as quark momenta inside one meson.

5 Higher channels, verification and the dynamical question

The opposite-order overlap suggested in Lin–Stanford App. F is

$$R_k = (-1)^k q^{d_1 d_2 + k(d_1+d_2) + k(k-1)/2}. \tag{19}$$

The authors report checks through $k = 25$, rather than an all-k proof of this formula. We use exact direct calculations for $k = 0, 1$ in Eq. (3); our further tests through $k = 4$ do not upgrade Eq. (19) to a theorem. Its exponent can be written

$$\tfrac{1}{2}\{D_k(D_k-1) - d_1(d_1-1) - d_2(d_2-1)\}. \tag{20}$$

This is an overlap between differently ordered states. Its modulus is below one at finite h and it must not be interpreted as an elastic scattering phase. For identical slots at the verified first grade, the physical form $G + G_\times$ retains the norm defect that produces the commutator contribution on page 2.

Independent checks. The saved validation contains 38 exact identities: first-grade primary/projection formulas, the complete six-letter Wick sum for F^2, Jacobi orthogonality, and relative Casimir eigenfunctions. Three additional symbolic differential identities verify the full two-variable S and B transformations and the Jacobi conjugation for general parameters. A separate recursive Gram implementation constructs primaries through $k = 4$ at

$$(h, d_1, d_2) = (0.05, 2, 2),\ (0.01, 2, 2),\ (0.02, 1.3, 2.1). \tag{21}$$

Computations at 85 and 115 decimal digits differ relatively by at most 2.5×10^{-78} in the recorded overlaps and projections. The maximum 115-digit overlap residual is below 1.8×10^{-108}. These are finite-degree numerical checks, not certified error bounds or evidence for uniform large-k limits.

The remaining physical input. In the tensor realization Q and B act only on P. Any suitable positive self-adjoint relative operator $\mathcal{M}_x^2$ therefore defines kinematically

$$P^+ = \mu P, \qquad P^- = \frac{\mathcal{M}_x^2}{\mu P}. \tag{22}$$

The algebra alone permits either continuous or discrete relative spectra. To choose $\mathcal{M}_x^2$ requires a microscopic spatial dictionary or independent dynamical Ward identities; channel counting does not choose it. The computed commutator decomposition supplies an interaction constraint that a candidate must reproduce.

There is also a distinction of sectors: two open X chords describe products of our *single* derivative singlet. They are useful for singlet interactions. The single-meson tower proposed by Miyashita–Sekino–Susskind instead calls for independent derivative fermion bilinears [5]. Their spectral matrix is the next direct microscopic target; the present relative index must not be relabeled as that tower.

6 Next calculation and reproducibility

We now have a positive total chord generator and an exact first relative-channel contribution to the noncommutativity of X. The next task is to enlarge the microscopic singlet operator family beyond X, so that particle-species information can be distinguished from time derivatives and products of the same singlet.

Start with the next derivative bilinears $\sum_i c_i^\dagger \operatorname{ad}_{H_s}^n(c_i)$ and their adjoints. Resolve their Hermitian components, conserved projections, and Liouvillian descendants before computing the remaining independent Gram and spectral matrix. The same disorder couplings must be retained inside the derivatives. Use existing complex-SYK chord/embedded-ensemble identities where available, followed by exact low-order checks. The criterion is a nonzero new independent spectral direction, not merely another correlator of X.

The channel result here remains a separate constraint for the eventual interaction map. We have not equated it with a spatial on-shell amplitude. The ordinary fixed-k, n contraction also leaves open singular ordering sectors, growing times, and the small-P domain of inverse translations. Those questions can be addressed when a concrete observable requires them, rather than replacing the spectral calculation.

Saved files. The stage `susskind-two-matter` contains the LaTeX source, Chinese derivation, literature map, numerical outputs and SHA-256 manifest. The numerical code is `two_matter_chords.py`. The independent checks are `check_two_matter_chords.py` and `check_two_matter_tensor.py`. The build script uses two `pdflatex` passes, and all six rendered pages are inspected. This stage is saved locally without a Git commit, push, or archive.

References

[1] H. W. Lin and D. Stanford, *A symmetry algebra in double-scaled SYK*, arXiv:2307.15725v1. Secs. 2.3, 3.3; Apps. B.2 and F.

[2] J. Xu, *On Chord Dynamics and Complexity Growth in Double-Scaled SYK*, arXiv:2411.04251v3, Sec. 3.2.

[3] H. T. Koelink and J. Van der Jeugt, *Convolutions for orthogonal polynomials from Lie and quantum algebra representations*, arXiv:q-alg/9607010, Corollary 3.6 and Remark 3.7.

[4] J. Van der Jeugt and R. Jagannathan, *Realizations of* $su(1,1)$ *and* $U_q(su(1,1))$ *and generating functions for orthogonal polynomials*, arXiv:math-ph/9807019, Eq. (2.10).

[5] S. Miyashita, Y. Sekino and L. Susskind, *Holograms and Standard Models*, arXiv:2607.05678v2, Secs. 4.2–4.3.